

E-Commerce Growth & Customer Analytics

Business Analytics & Power BI Project Report

Interactive dashboard covering Sales, Customer Behavior, Conversion, Product Performance and Marketing.

Portfolio project documentation



1. Project Overview

This project is an e-commerce business analytics solution built in Power BI. It translates transaction, customer, clickstream, product, review and traffic-source data into an interactive decision-support dashboard. The dashboard moves from overall business performance to customer behavior, conversion leakage, product performance and acquisition-channel effectiveness.

2. Domain & Business Understanding

Domain: E-commerce / Digital Commerce Analytics

E-commerce businesses collect data across visits, customer activity, product interactions, orders, reviews and acquisition channels. Business teams need to know what is selling, who is buying, where customers leave the purchase journey, which products drive revenue, and which traffic sources create valuable outcomes.

Area	Business need
Sales & Revenue	Monitor revenue, orders, AOV, profit and growth.
Customer	Understand customer type, engagement, purchasing and feedback.
Conversion	Identify movement and drop-offs from browsing to purchase.
Product	Identify high-performing products and categories.
Marketing	Compare traffic sources on traffic, conversion and revenue.

3. Problem Statement

How can an e-commerce company use transactional, customer, behavioral and traffic-source data to monitor business performance, understand customer behavior, identify conversion drop-offs, evaluate product and category performance, and determine which acquisition sources contribute most to growth and revenue?

4. Project Objectives

- Build an interactive Power BI dashboard for management-level monitoring.
- Measure sales and revenue growth using current-period and previous-year analysis.
- Understand customer engagement, purchase behavior and review sentiment.
- Measure the journey from session to product interaction, cart and purchase.
- Identify high-performing products and categories.
- Evaluate traffic sources by volume, conversion and revenue efficiency.
- Demonstrate Power BI skills including DAX, time intelligence, bookmarks, buttons, slicers, field parameters and interactive navigation.

5. Business Questions

- What is the overall sales and revenue performance?
- How does current revenue compare with the previous year?
- Are orders and AOV improving?
- How do customers behave and engage with the platform?
- How do new and repeat customers differ in purchasing behavior?
- How are customer reviews distributed across positive, neutral and negative categories?
- How many sessions progress from product view to add-to-cart to purchase?
- Where is the largest funnel drop-off?
- Which devices, categories and traffic sources have stronger conversion?
- Which products and categories contribute most to revenue and sales?
- Which traffic sources generate the strongest traffic, conversion and revenue outcomes?

6. Data Understanding

The working Power BI model uses the following business entities/tables referenced during development.

Table	Purpose	Fields referenced
orders	Completed transaction information	order_id, customer_id, order_time/order_date, total_usd
sessions	Visit/session information	session_id, customer_id, session start time
events	Clickstream and funnel events	session_id, event_type
products	Product master data	product_id, product name, category
order_items	Product-level order detail	order_id, product_id, quantity
customers	Customer master data	customer_id, signup_date
reviews	Customer feedback	review, rating / review text
Date	Time dimension	Date, Year, Month, Quarter, Year-Month, Week

7. Data Model & Analytical Logic

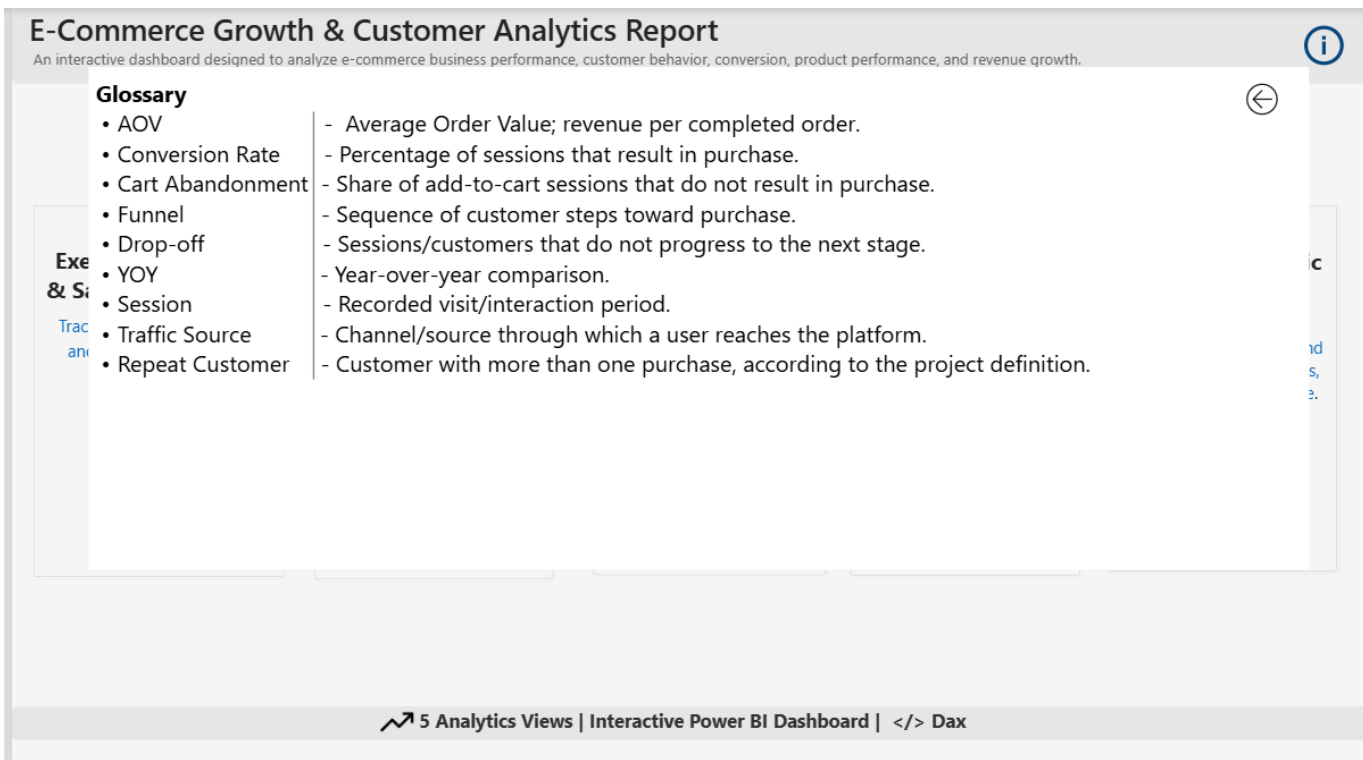
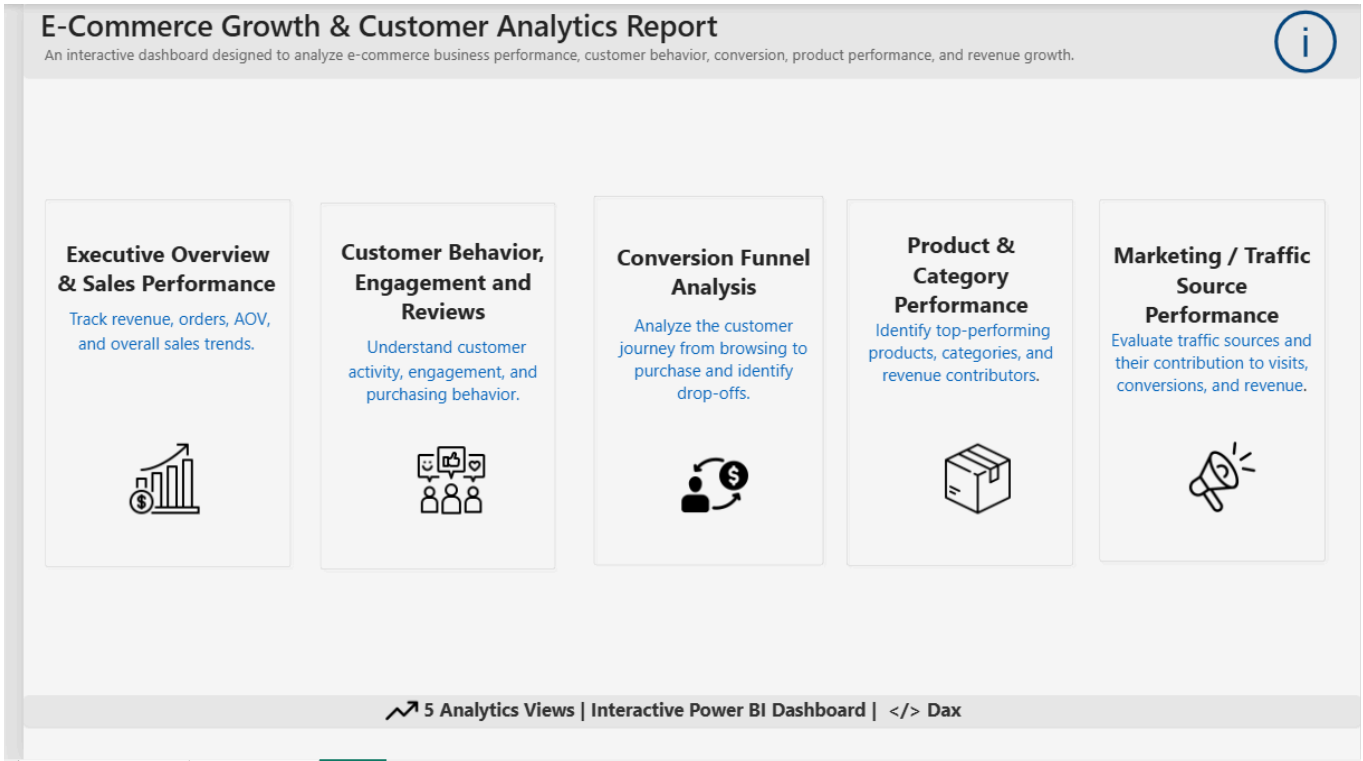
- Created a dedicated Date table and connected the date-only order date to the order table.
- Used the Products → Order Items → Orders relationship path instead of creating an artificial direct Products → Orders relationship.
- Used event_type and session_id for funnel calculations.
- Calculated customer segmentation at customer level rather than treating each order row as a customer.
- Used DAX measures for reusable business KPIs and time-intelligence calculations.
- Used a generalized review scale: Excellent/Good = Positive; Okay Overall = Neutral; Not Buy Again/Very Poor = Negative.

8. Dashboard Story (Tab by Tab)

Landing Page / Navigation

Starts the business story and routes users to five analytical views. The landing page is designed to demonstrate clean navigation using buttons/bookmarks.

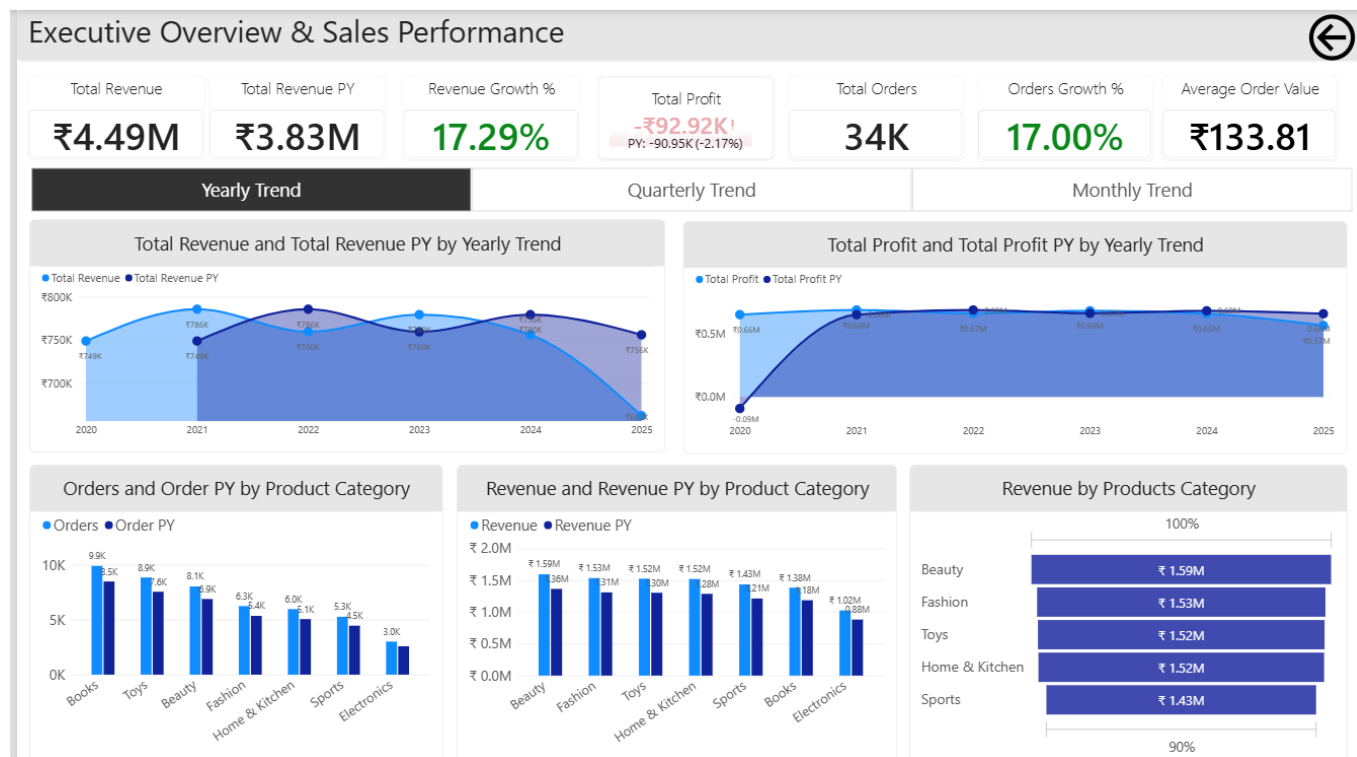
Business role: Provide a clear executive entry point instead of presenting disconnected charts.



Sales & Revenue Performance

Answers: Is revenue growing? How does revenue compare with the previous year? Are orders and AOV improving? How is profit changing?

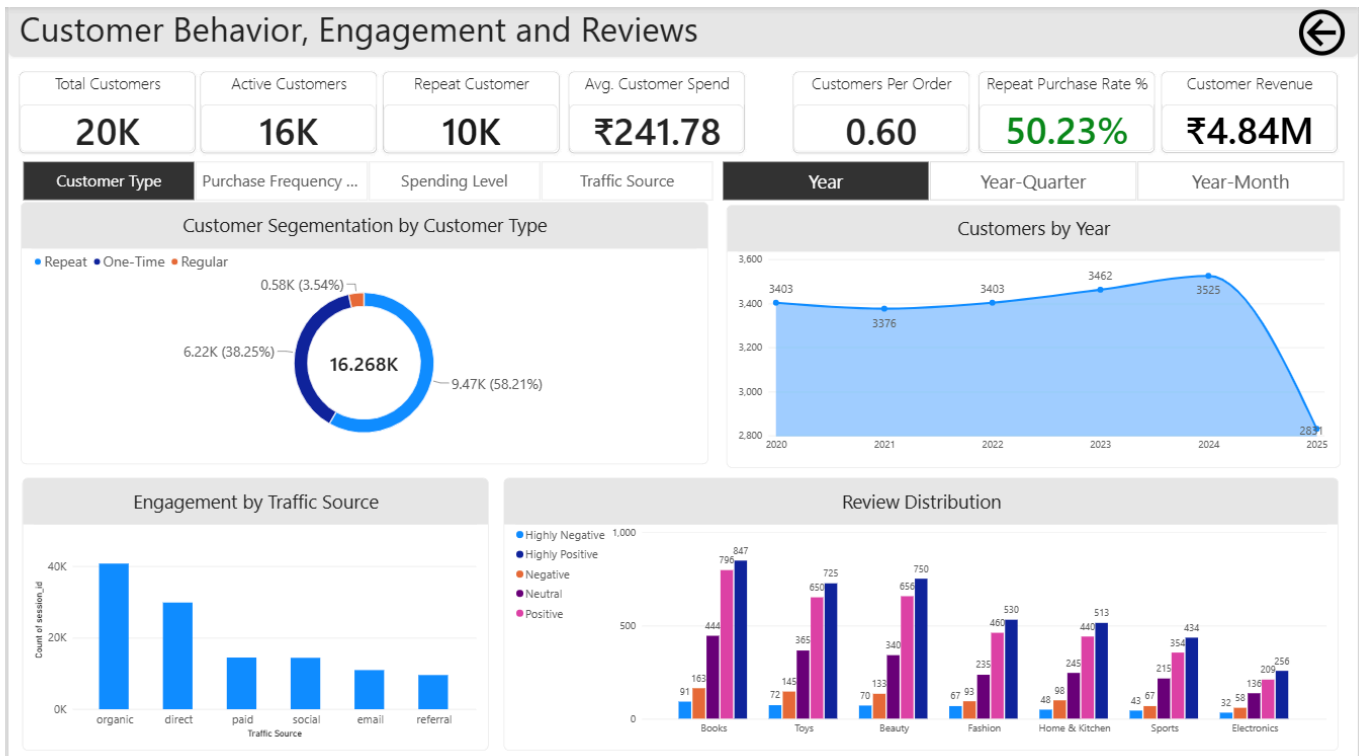
KPIs: Total Revenue, Total Revenue PY, Revenue Growth YOY%, Total Profit, Total Orders, Order Growth %, AOV. Visuals Cover revenue/PY trends, profit/PY trends, orders and revenue by category, and top-5 categories/products.



Customer Behavior, Engagement & Reviews

Answers: Who are the customers? How do they engage? What are their purchase frequency and spending levels? How does behavior differ by device/customer type/traffic source? What does feedback look like?

Analysis includes customer activity, new vs repeat customers, device behavior, traffic source, purchase frequency, spending level and review sentiment.

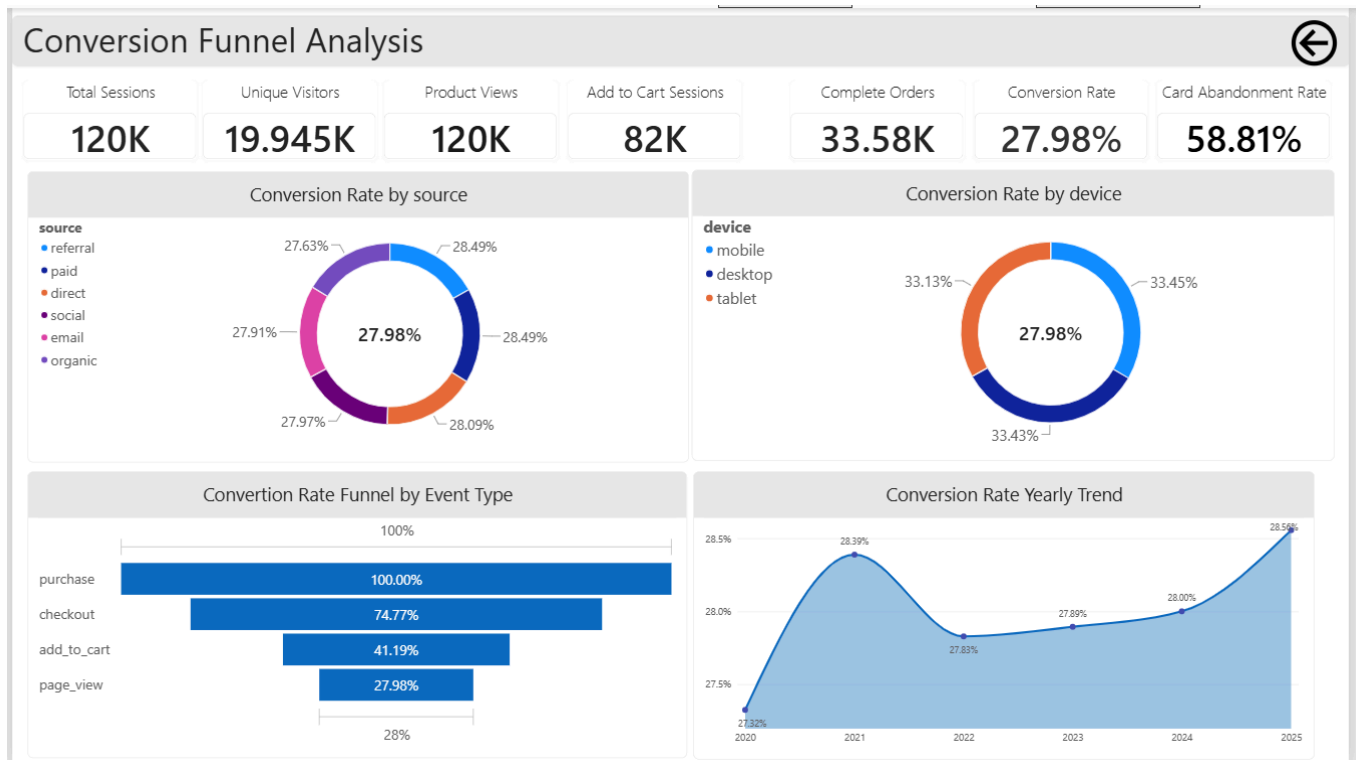


Conversion Funnel Analysis

Answers: How does a session progress from product view to add-to-cart to purchase? Where does the largest drop-off happen? Which category, device or source converts better?

KPIs: Total Sessions, Unique Visitors, Product Views, Add-to-Cart Sessions, Completed Orders, Conversion Rate and Cart Abandonment Rate.

Supporting analysis includes conversion trends, conversion by category/device/source and funnel drop-off.

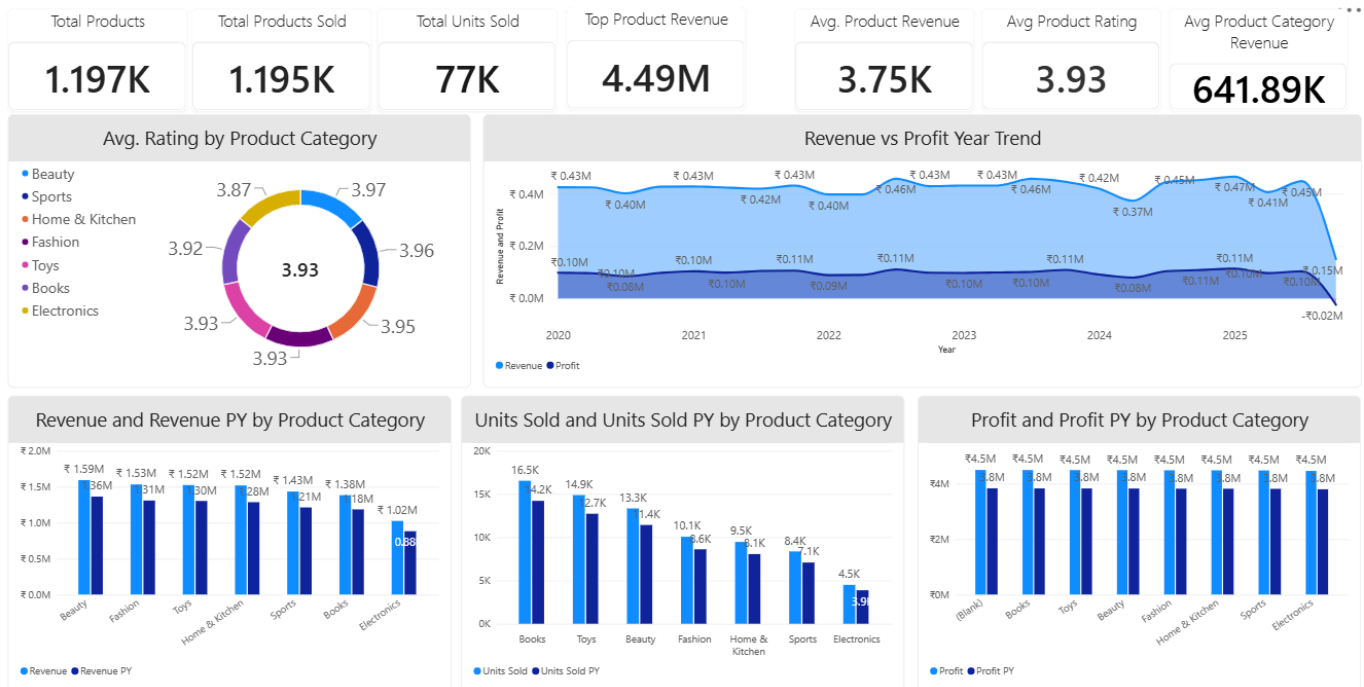


Product & Category Performance

Answers: Which products/categories drive revenue, orders and units? Which products are the strongest contributors? How does performance change over time?

Uses product/category rankings, revenue, profit, units, orders, margin and matrix views. Field parameters/bookmarks allow dynamic analytical views.

Product & Category Performance

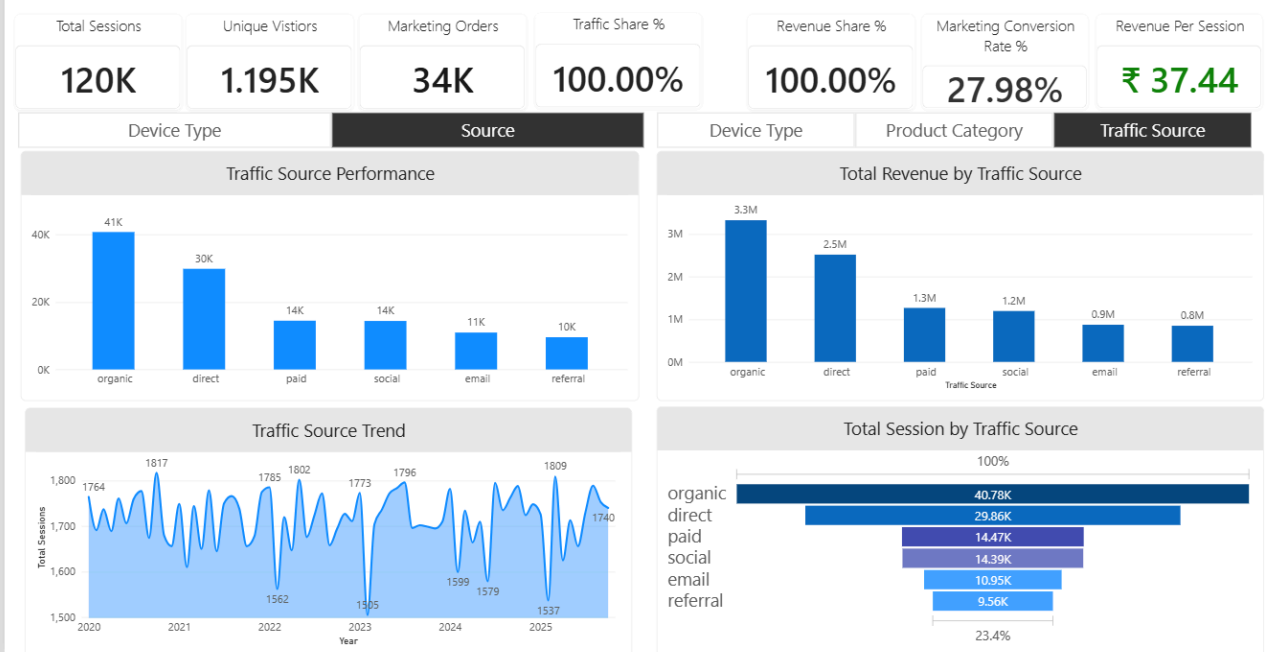


Marketing / Traffic Source Performance

Answers: Which sources bring traffic? Which convert? Which generates revenue? Which sources provide the strongest revenue efficiency?

Focuses on marketing sessions, marketing conversion, marketing revenue, revenue per session, traffic share, revenue share, source trends and funnel contribution.

Marketing / Traffic Source Performance



9. KPI Definitions

KPI	Definition
Total Revenue	Total sales value from orders.
Total Revenue PY	Comparable previous-year revenue.
Revenue Growth YOY %	Percentage change in revenue versus the previous year.
Total Profit	Revenue less the relevant cost basis used in the model.
Total Orders	Distinct completed orders.
Order Growth %	Percentage change in orders versus the previous year.
AOV	Average order value = revenue / orders.
Conversion Rate	Purchase sessions divided by total sessions.
Cart Abandonment Rate	1 – purchase sessions / add-to-cart sessions.
Revenue per Session	Revenue divided by sessions.
Traffic Share %	Selected source sessions as a share of total sessions.
Revenue Share %	Selected source revenue as a share of total revenue.

10. Technical Skills Demonstrated

Skill	Application
Power BI	Data model, visuals, interactions and dashboard storytelling
DAX	KPIs, time intelligence, conversion, segmentation and business measures
Power Query	Data preparation and transformations where required
Bookmarks & Buttons	Landing-page navigation and interactive views
Field Parameters	Dynamic category/metric analysis
Data Modeling	Date dimension and relationship-based analysis
Business Analytics	Translating data into business questions and actions

11. Business Insights — Finalize From the Dashboard

This section should contain actual observations from the finished dashboard.

- Revenue changed by **17.29%** YoY during the analyzed period.
- The largest funnel drop-off occurs between **checkout** and **add to cart**, at **33.58%**.
- **Beauty** is the highest-revenue category/product.
- **Referral and Paid** is the strongest-converting traffic source at **28.49 %**.
- Repeat customers show **50.23%** purchase rate compared with new customers.
- Review sentiment is predominantly **positive** (or **highly positive**), while the largest negative category is **Books**.

12. Recommendations — Tie Them to Evidence

Observed pattern	Potential business action
High traffic + low conversion	Investigate landing/product experience, pricing, targeting and checkout friction.
High add-to-cart + low purchase	Investigate checkout, shipping/payment friction and trust signals.

High-revenue category	Protect availability and test merchandising/cross-sell opportunities.
High sessions + low revenue/session	Review traffic quality and acquisition targeting.
High conversion + low traffic	Evaluate controlled scaling of the source.
Negative review concentration	Investigate product/service issues and prioritize corrective actions.

13. Glossary

Term	Meaning
AOV	Average Order Value; revenue per completed order.
Conversion Rate	Percentage of sessions that result in purchase.
Cart Abandonment	Share of add-to-cart sessions that do not result in purchase.
Funnel	Sequence of customer steps toward purchase.
Drop-off	Sessions/customers that do not progress to the next stage.
YOY	Year-over-year comparison.
Session	Recorded visit/interaction period.
Traffic Source	Channel/source through which a user reaches the platform.
Repeat Customer	Customer with more than one purchase, according to the project definition.
Field Parameter	Power BI feature used to dynamically switch fields/measures in visuals.
DAX	Data Analysis Expressions, Power BI's formula language.

The analytical story follows the decisions a business team would make: first understand overall performance, then understand customers, identify where customers are lost in the purchase journey, determine which products drive results, and finally evaluate which acquisition sources create valuable business outcomes. The dashboard therefore demonstrates both Power BI implementation skills and business-oriented analytical thinking.

14. Video Walkthrough

 E-Commerce Dashboard VideoWalkthrough.mp4